

# General CS Problem-Solving

The lexicon, and the two evidence channels the one name covers

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The index question is one line and it is the hardest one in the cluster.

*“what mental operations does it take to solve generic problems in computer science?”*

website, Problem-Solving, the three settings

The page answers it in the register of a coding study: a recorded session, three coders, a consensus track, an agreement figure that is deliberately not flattering. The work on disk answers it in the register of a population study: thirteen million submissions to two online judges, no transcripts, no verbalisation, and a research question about what survives when the rich record is thrown away. Both are this project. Neither page mentions the other.

That is not a contradiction, it is the design. The two channels are put against each other on purpose, and the third research question is the bridge: which components of the framework survive reduction from a rich record  $R$  to a CodeNet-style  $g(R)$ . This file keeps both channels in view for every term, because most of the vocabulary means something slightly different in each.

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# 1 Active

## 1.1 The object

### session, and solve

*“A session isn’t a list.”*

website, General CS Problem-Solving, solving as a graph

**Definition:** One person working one problem from first reading to whatever end it reaches. A solve is the same stretch named by its outcome. Drawn as a graph rather than a sequence, so that the branching, the loops back, and the step that turned out to be the hinge are all visible.

**Distinguishes it from:** The episode, which is one bounded stretch inside a session, and the trajectory, which is what a session looks like after the observability is reduced to submissions. One session holds many episodes. One trajectory may hold a session that no one recorded.

**Operational test:** Can the record show two things spawned by one step at the same time? Reading the spec spawns both a plan and the memory of a similar bug. If the record cannot represent that, it is a list and the graph claim is decoration.

**In this project:** The rich channel records the session as a transcript, typed, uploaded, or recorded in the browser and transcribed. The reduced channel identifies the same unit as a user and problem trajectory, and there are 6,764,563 of them, of which 2,532,957 hold more than one submission.

### mental operation, in this project

**Definition:** The cluster definition holds: a discrete act of thought that transforms one mental state into another, held at a stated grain. This project fixes the grain by writing a codebook, twelve object-level codes and eight regulatory ones, each with positive evidence cues and an explicit exclusion.

**Distinguishes it from:** The hyperedge of the core framework, which is the formal object and carries a source set, a target set, a relation type, and a confidence. The codebook code is what a coder can apply. The hyperedge is what the framework quantifies over. They are two representations of one construct and the project needs both, but only one of them can be assigned by a human reading a transcript.

**Operational test:** Does the label survive its own exclusion clause? Each code carries one. Reading is not Represent without evidence of a constructed representation. Keystrokes are not Translate, because they do not reveal what plan is being implemented. Judge acceptance is not Verify. If the evidence you have only clears the exclusion, the label is not supported.

**In this project:** The construction principle is stated plainly: “No validated universal taxonomy spans algorithm design, program comprehension, debugging, implementation, and self-regulation.” So the codebook is hybrid and hierarchical, general theory supplying the parents and programming studies the domain-specific children, and “Parent labels are retained only when independent coders can distinguish them at the selected temporal grain.”

### Seven operations on the page, twenty codes in the codebook

The page lists seven: “Reading, decomposing, recalling something they’ve seen before, planning, implementing, testing, revising.”

The codebook has twelve object-level codes, O1 Represent, O2 Recognize or retrieve, O3 Relate or analogize, O4 Decompose or compose, O5 Design or derive, O6 Translate or implement, O7 Predict or simulate, O8 Test or probe, O9 Diagnose or localize, O10 Repair, O11 Verify or explain, O12 Reuse or abstract, and eight regulatory ones above them, R1 through R8.

Six of the page’s seven land in the object-level set. One does not. Planning is R1, goal and resource planning, and it sits in the regulatory layer by construction, above the object-level work rather than mixed with it. So the page’s list of seven crosses the exact boundary the codebook exists to draw.

The page is not wrong; it is the informal statement of the question, and the informal statement is where planning naturally sits next to implementing. The codebook is the answer, and its first act is to separate them. Anywhere the two appear together, say which one is being used.

### coder, and the consensus track

*“Three coders labelling the same session, segment by segment, with the consensus track underneath.”*

website, General CS Problem-Solving, can observers agree on it

**Definition:** An annotator labelling a session against a shared scheme, independently of the others. The consensus track is the derived sequence beneath the three, solid where all three agree and faded where they do not.

**Distinguishes it from:** An adjudicator, who resolves disagreements. This project has no adjudication step in the reporting path: where the three split the consensus fades rather than picking a winner, and the split is retained as a result.

**Operational test:** Did the coder see another coder’s labels before submitting? If yes the agreement figure means nothing.

**In this project:** Three coders per session in the animation. Optional item identifiers let segments line up across coders, which is what makes segment-level comparison possible at all.

### segment, and the boundary problem

*“Naming those operations is easy. Getting independent observers to agree on where one ends and the next begins is not”*

website, General CS Problem-Solving, the question

**Definition:** The stretch of a session that one label covers. Segmentation is the assignment of boundaries; coding is the assignment of labels to what segmentation produced.

**Distinguishes it from:** The label. These are two failure modes and they are usually reported as one number. Two coders can agree completely on the vocabulary and disagree completely on where a stretch begins.

**Operational test:** Can a coder mark a boundary without choosing a label for what follows it? If the instrument forces the two together, the two error sources cannot be told apart afterwards.

**In this project:** Labels overlap by design. Recognising that a test falsifies an approach and

replacing the algorithm spans R2 monitoring, R4 replanning, O9 diagnosis, O5 design, and O6 implementation at nested spans, so a segment is not a partition. Macro spans hold strategy cycles, meso spans hold one operation, and micro cues are the verbal, behavioural or artifact evidence beneath.

## 1.2 The standard

### construct, and reliability

*“that agreement is the whole ballgame, a construct nobody can code reliably isn’t a construct”*

website, General CS Problem-Solving, the question

**Definition:** The project’s admission rule for a label. A name that independent coders cannot apply the same way is not admitted to the analysis, no matter how good the name is.

**Distinguishes it from:** An exhaustive account of what a person is doing. The closing sentence of the codebook says what the output is meant to be: “a validated working ontology of recoverable programming operations, not an exhaustive inventory of cognition.”

**Operational test:** Report agreement before merging, and again after. If only the second number exists, the scheme was tuned to the number that reports it.

**In this project:** The validation cycle is written out: open coding on a diverse pilot sample, merge the labels coders cannot distinguish, add sublabels only when they have distinctive evidence cues, then test the revised codebook on held-out episodes and tasks. Reporting includes per-label prevalence and reliability, agreement before and after merging, cross-task stability, negative cases, and a version history. Sublabels enter primary analysis only after the parent labels are reliable.

### stage-level percent agreement, and Cohen’s kappa

**Definition:** The two agreement figures the instrument reports between annotators, exportable as a table.

**Distinguishes it from:** Each other, and the difference is the whole reason both are reported. Percent agreement counts matches. Kappa discounts the matches chance would produce, so a scheme with one dominant label can post high agreement and near-zero kappa.

**Operational test:** Is one label carrying most of the segments? If so, quote kappa, and quote percent agreement only next to it.

**In this project:** The running agreement figure on the page is described as “deliberately not flattering”. Astrolabe computes both and exports them.

### evidence strength

**Definition:** Four grades on the support for a single label. A, direct: concurrent verbalisation names the operation and is aligned in time with the behaviour. B, triangulated: behaviour, an artifact or test change, and the outcome jointly support it. C, inferential: the trace is consistent with the operation and permits plausible alternatives. D, unsupported: the label rests only on a verdict, a delay, or static code.

**Distinguishes it from:** Coder agreement. Three coders can agree perfectly on a label whose evidence is grade C, and that is the common case in the reduced channel.

**Operational test:** Strip the verbalisation from the record and re-read the segment. If the label still follows, it was not A.

**In this project:** Strong claims about operations require A or B. The reduced channel will often support only C and the rule is that it must carry that uncertainty rather than round it up.

### 1.3 The reduced channel

#### bounded phenomenon

*“The immediate study concerns closed online-judge programming episodes from AIZU and AtCoder, not programming or CS problem solving generally.”*

one-pager, research statement

**Definition:** The population the empirical claims cover: timestamped submissions made by one anonymised user to one judged task.

**Distinguishes it from:** The project’s title. “The population is narrower than programming, software engineering, computer-science learning, or computer-science problem solving in general.” The website page is named General CS Problem-Solving and the study is not general. The generality is the ambition; the population is the claim.

**Operational test:** Does the dataset identify professional status, expertise, learning history, collaboration, local development activity, or use of external resources? None of them, so none of them may appear in a conclusion.

**In this project:** The non-claims are enumerated rather than implied. The study does not treat submissions, verdicts, time gaps, or syntax-tree changes as mental operations; does not claim accepted programs are unique, optimal, or proof-correct; does not infer expertise or learning from anonymised identifiers; does not treat verbal reports as infallible cognition; and does not generalise to professional, collaborative, open-ended, or non-programming work.

#### reduction, $g(R)$

**Definition:** The operation that takes a rich record  $R$ , with editor events, keystrokes, outcomes and think-aloud, and returns what a submission archive would have kept. The third research question asks which framework components survive it.

**Distinguishes it from:** Sampling, which keeps fewer records of the same kind. Reduction keeps every record and removes channels from each, which is a different loss and it falls unevenly across the codebook.

**Operational test:** Code the rich record, then code the reduced record blind, then compare. The design says the reconstruction from the reduced trace is blinded and disagreement is retained.

**In this project:** The reporting categories are fixed in advance: supported, ambiguous, and unidentified. No channel is treated as ground truth, including the think-aloud.

#### trajectory

**Definition:** The ordered submissions of one user to one problem, with source snapshots, verdicts, timestamps, language, runtime, memory and code size.

**Distinguishes it from:** The solve. A trajectory is what the judge saw. The solve includes everything that happened between submissions, which is most of it. The dataset was not designed or validated as a representation of cognitive process, and the project says so about its own primary source.

**In this project:** 13,916,868 submissions, 7,460,588 accepted artifacts, 4,053 problems, 55 languages, 6,764,563 user and problem trajectories. 2,532,957 of them, 37.44%, hold multiple submissions, and 2,328,506, 34.42%, hold multiple submissions in one language. Adequate for population-scale revision analysis and not for unqualified claims about mental content.

## concept layer

**Definition:** The versioned external backbone of concepts, annotated for task relevance, with evidence drawn from problem statements and programs, typed relations between concepts, and activation recorded per episode.

**Distinguishes it from:** Anything the dataset supplies. It does not supply this. The rating, tags and complexity fields are empty or unused, so the concept layer is hybrid by necessity, built from curricula and algorithm sources and double-annotated.

**Operational test:** Is an accepted program standing in for a solution path? The rule is that accepted artifacts are never substituted for mental hyperpaths, and program graphs count as structural evidence rather than as prerequisites.

**In this project:** Three of the five primary empirical claims, trajectory structure, task and difficulty description, and artifact and performance structure, are reproducible from the dataset alone. The concept graph is not one of them.

## 1.4 The instrument

### Astrolabe

*“a local research app built for this work”*

website, General CS Problem-Solving, the instrument

**Definition:** The annotation instrument the rich channel runs on. Takes in transcripts, typed, uploaded, or recorded in the browser and transcribed; lets several annotators code one session against a shared scheme; reports stage-level percent agreement and Cohen’s kappa; exports the agreement report and the full annotation bundle.

**Distinguishes it from:** A qualitative coding suite. Two commitments separate it. “It runs entirely on your own machine.” And it never writes to the source material: it reads one snapshot on first load, keeps its own state in a separate database, and hands changes back as pasteable text the researcher applies. You can keep editing the original while it runs.

**Operational test:** Open the source file in another editor while the app is running and change it. If the app has to be told, or overwrites the change, the read-only guarantee is not in force.

**In this project:** It also carries the question-card structure the project is organised around, hypergraph overlays for grouping operations at different scales, eleven graph scores that shade the nodes live, six export formats, and reusable prompt scaffolds for the case where a language model proposes candidate operations or concepts.

## question card, and origin edge

**Definition:** A qcard is one question parsed from the source document and carried as a node. Structural edges run parent to child and record decomposition. Origin edges, drawn dashed, record which question spawned which.

**Distinguishes it from:** Chronological order. The timeline view lays the cards out in the topological order of the origin edges, which is not the order in which they were written, and the difference between those two orders is itself readable.

**Operational test:** Would removing the parent make the child unanswerable, or merely unmotivated? The first is structural, the second is origin.

**In this project:** These are the observable stand-in for the framework’s question nodes. The framework’s own four relation types, compositional, causal, co-occurrence and dependency, are a finer vocabulary than the two edge kinds the instrument records, and the mapping between them is not yet fixed.

## hinge

*“which step turned out to be the hinge”*

website, General CS Problem-Solving, solving as a graph

**Definition:** The step in a solve that the rest of the solve turned on.

**Distinguishes it from:** The longest step, and the step where the solver was most stuck. The instrument computes it from graph structure rather than from duration or affect, using eleven scores over blocking, decisiveness, relevance and their combinations, and shades the highest-scoring card.

**Operational test:** Remove the step and ask whether the goal question is still reachable. That is the graph-theoretic version, and it is the reason criticality in the framework is defined as value of information on the dependency graph rather than as a property of the step itself.

**In this project:** The claim on the artifact page is a claim about method: which subproblem was the hinge becomes visible rather than argued.

## 2 Working

### snapshot

**Definition:** The single read of the source material Astrolabe takes on first load, from which all its own state is derived.

**Distinguishes it from:** The code snapshot in the reduced channel, which is the submitted source of one submission. Same word, two channels, two objects. Named in both places in this project and worth disambiguating on sight.

### shared scheme

**Definition:** The label set several annotators code the same session against.

**Distinguishes it from:** The codebook, which is the scheme plus the evidence cues, the exclusions,

the temporal hierarchy, and the version history. A scheme is a list. A codebook is a list you can hand to someone else.

### sublabel

**Definition:** A finer code beneath a parent: working backward, case splitting, complexity forcing, boundary testing, minimal counterexamples, forward and backward localisation, slicing, syntax or boundary repair, performance repair, algorithm replacement.

**Distinguishes it from:** The parent code. Sublabels are exploratory and enter primary analysis only after parent-label reliability is established, which is the grain decision applied a second time one level down.

### process mining

**Definition:** Discovery of process structure from timestamped event logs.

**Distinguishes it from:** What this project does with the same logs. Discovered sequences carry no intrinsic cognitive interpretation, so process discovery is used here as a baseline the construct account has to beat, not as the account.

### learning analytics

**Definition:** Prediction of learning or performance from submissions, errors, test results and educational records.

**Distinguishes it from:** The target here. Predicting whether someone will succeed does not identify the questions, hypotheses, or operations that organise one episode, and the project's positioning says so in those words.

## 3 Peripheral

### Polya, Schoenfeld, Newell and Simon, GOMS, ACT-R

The inherited scaffolding behind the codebook. Polya's understand, plan, execute and look back is used as an episode scaffold and not as a chronology. Schoenfeld separates resources, heuristics, control and beliefs, which is why possessing a strategy is treated as different from selecting it. Newell and Simon supply operators over problem states. GOMS and ACT-R are the demonstration that operations occupy several temporal scales at once, which is the grain problem in its original form.

### debugging decomposition

Comprehension, testing, fault localisation, repair, with forward reasoning from the code toward its effects and backward reasoning from a symptom toward its cause. The clearest existing catalogue of domain-specific operations, and the reason the codebook's middle is denser than its ends.

### program slicing

Reduction of a program to just the statements that can affect a chosen value. The mechanised form of the backward tracing a debugger does by hand, and a candidate artifact-side cue for O9.

## overly fine metacognitive schemes

The recorded failure the coarse grain is a response to: elaborate metacognitive taxonomies produce poor correspondence between raters and become useful only after the categories are restricted and recombined.

## 4 Where it connects

*“If mental operations can be identified reliably, then a system that watches a work session can say something useful about where someone is stuck rather than only that they are”*

website, General CS Problem-Solving, where it connects

The one-pager states the same dependency from the assistance side, and it is the sentence that says why this project sits underneath the rest of the cluster rather than beside it.

*“AI cannot complement a human solver optimally if it represents only whether the final program passes.”*

one-pager, research statement

The order is fixed by that dependency and the project holds to it. An intervention enters the immediate study only if the validation identifies an assistance target that recurs, is recoverable, and matters. Nothing is built until then, which is a discipline the assistive half of the portfolio does not otherwise impose on itself.

## 5 What this project leaves open

The page’s list of seven operations is a sentence, not a scheme, and the codebook that turns it into one has not yet been through the pilot that merges its confusable labels. Until it has, the twenty codes are candidates and their count will change.

The two channels have not been joined. The reduction question is written, the reporting categories are fixed, and the blinded comparison is designed, but the rich record it needs is the recorded, coded sessions the website page shows, and the population it needs is the submission archive. Nothing yet runs both ends.

The mapping between the instrument’s two edge kinds, structural and origin, and the framework’s four relation types is not written down anywhere. That is a small gap and it will become a large one the moment a coded session is exported into the hypergraph and the edge labels have to be chosen.

The publication line on the page is one sentence: in preparation, with the instrument described under Artifacts. That is accurate and it is the reason this file quotes the design rather than results.

## 6 Sources

- Sentences marked *website* are from the General CS Problem-Solving page under `3-year-agenda/mental-world-modeling/ problem-solving/`, from the Problem-Solving index above it, or from the Astrolabe page under `content/artifacts/`.

- Sentences marked *one-pager* and every dataset number are from the CHI 2027 research plan, Notes/Project One-Pager in the DesignTheGame folder [3] 2026 Ana\_concensus\_on\_mental\_operations\_in\_cs\_ problem\_solving, sections final\_research\_statement, final\_research\_design, research\_positioning and mental\_operation\_codebook.
- The codebook's twenty codes, their evidence cues, their exclusions, the four evidence grades and the validation cycle are from sections/mental\_operation\_codebook.tex.
- The formal question, operation, relation type and criticality definitions are from core idea/main.tex, and the account of grain and of the regulatory layer from core idea/sections/on\_mental\_operations.tex.
- Shared cluster terms, mental operation, grain, the control layer, reliability, disagreement as finding, elicitation, are defined once in mwm\_shared.tex and are not repeated here.